DERIVED TENSOR PRODUCTS STAY IN D_QCOH

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 652-662, immediately following the statement at 644-650.

Editorial note. The pullback-tensor commutation is the span's own step; the two tensor functors live over different rings, which the terminal lemma needs.

Allowed standard prerequisites (editorial).

- The covering criterion for quasi-coherence. Lemma lemma-check-quasi-coherence-on-covering of this chapter, cited at proof lines 657-661. Reduces the claim to the scheme case via the surjective etale cover.
- Scheme case. Derived Categories of Schemes, Lemma perfect-lemma-quasi-coherence-tensor-product, cited at proof lines 658-661. Terminal input: derived tensors of D_QCoh objects on a scheme stay in D_QCoh.

Source reference. Original lemma 08F2: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 535–535. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08F5: focused-seeds/geometry/S9-perf-quasi-coherence-tensor-product/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. For objects K, L of $D_{QCoh}(\mathcal{O}_X)$ the derived tensor product $K \otimes^{\mathbf{L}} L$ is in $D_{QCoh}(\mathcal{O}_X)$.*

Proof. Let $\varphi : U \to X$ be a surjective étale morphism from a scheme U. Since $\varphi^*(K \otimes^{\mathbf{L}}_{\mathcal{O}_X} L) = \varphi^* K \otimes^{\mathbf{L}}_{\mathcal{O}_U} \varphi^* L$ we see from Lemma 08F2 that this follows from the case of schemes which is Derived Categories of Schemes, Lemma 08F5. □

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.